

ANTI-PROXY CLASSROOM ATTENDANCE SYSTEM

Student Names & Roll nos:

- Vasudev V – 70
- Sulfa Saji – 63
- Susan Saji – 64
- Meera Krishna S – 75

Guide Name : Er. Gayatri J L

Department: Computer Science & Engineering
College: Saintgits College of Engineering
Academic Year: 2023–2027

Introduction

- Attendance monitoring plays a vital role in academic evaluation
- Traditional attendance systems are prone to proxy attendance
- Manual and basic digital systems are time-consuming and inefficient
- AI-based face recognition provides a secure and automated solution

Problem Statement

- Proxy attendance compromises academic integrity
- Manual roll calls waste classroom time
- Existing systems lack identity verification
- Need for a reliable, automated, and secure attendance system

Objectives

- To analyze limitations of existing attendance systems
- To design an AI-based attendance solution
- To implement face detection and recognition
- To prevent proxy attendance effectively
- To improve accuracy and efficiency

Literature Survey

AUTHORS	YEAR	TITLE	TECHNIQUE USED	LIMITATION
Abhishek Kumar, Ankit Gupta, Ritu Sharma	2019	Automated Attendance System Using Face Recognition	CNN-based face recognition	Struggles in real-time crowded classrooms
Md. Asif Iqbal, Md. Rifat Hossain, Md. Mahmudul Hasan	2020	Smart Attendance System Using Deep Learning	Deep CNN for face recognition	Sensitive to lighting variations
Rohit Kumar, Saurabh Singh, Neha Verma	2021	Face Recognition Based Attendance System Using Deep Learning	Deep learning embeddings + similarity matching	Computationally expensive
S. Harish, R. Karthik, P. Sathish Kumar	2021	Real-Time Face Recognition Based Attendance System	CNN + OpenCV	Lower accuracy in multi-face scenarios
Pradeep Kumar, S. Balaji, M. Prakash	2022	YOLO-Based Real-Time Attendance Monitoring System	YOLO for face detection + deep learning recognition	Requires high-quality camera setup

Proposed System

- AI-based face recognition attendance system
- Real-time face detection using camera with YOLO
- Three face images are captured per student during each class period
- Captured images are mapped to the corresponding period
- Face embeddings used for identity verification
- Attendance stored automatically in database

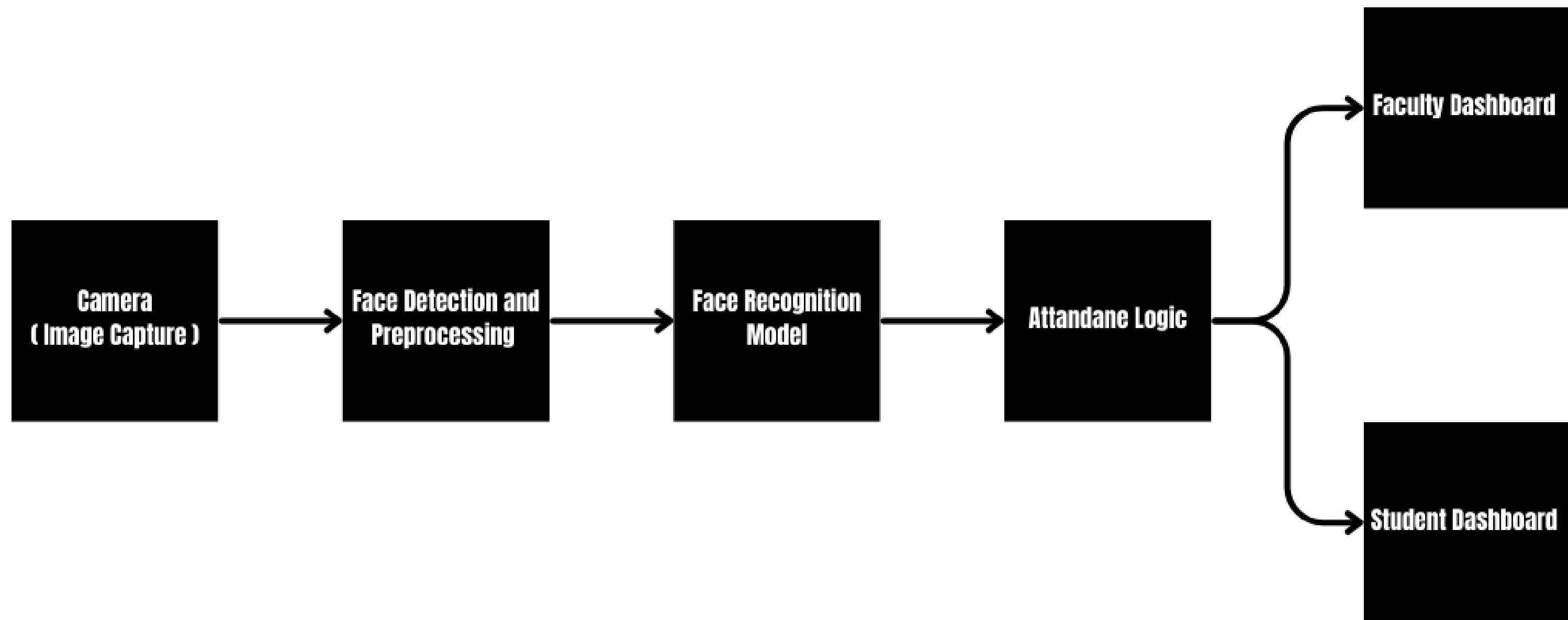
Advantages:

- Eliminates proxy attendance
- Fully automated
- High accuracy and transparency
- Reduces false attendance using multi-image verification

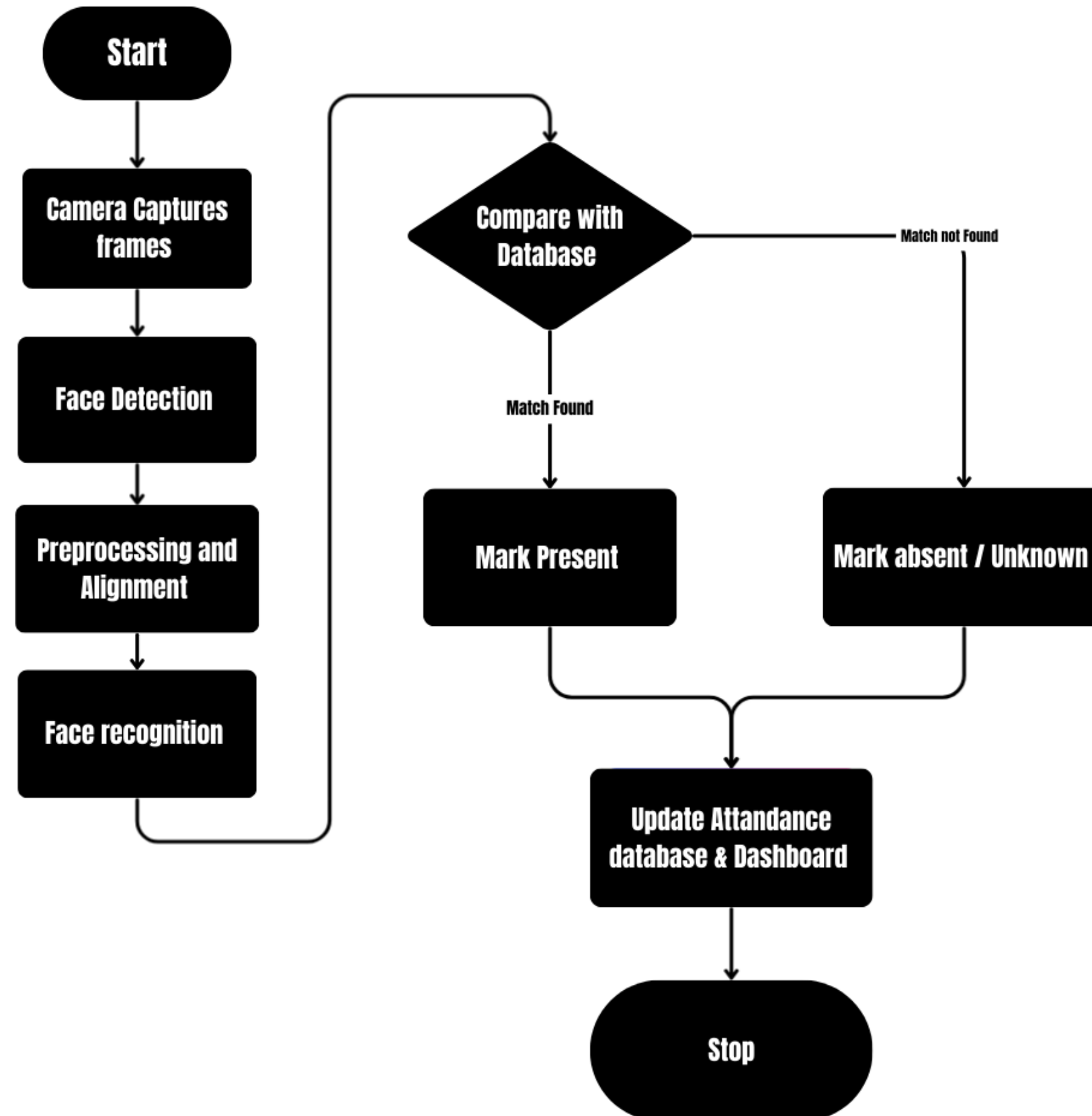
System Architecture

- Camera captures 3 images per student during each class period
- YOLO-based face detection applied on captured images
- Preprocessing and face alignment performed
- Face embedding generation using FaceNet
- Similarity matching using cosine similarity
- Attendance mapped to the corresponding period
- Attendance marked and stored in database
- Attendance details displayed in student and faculty dashboards

System Architecture Diagram



Flow Diagram :



Technologies And Tools Used

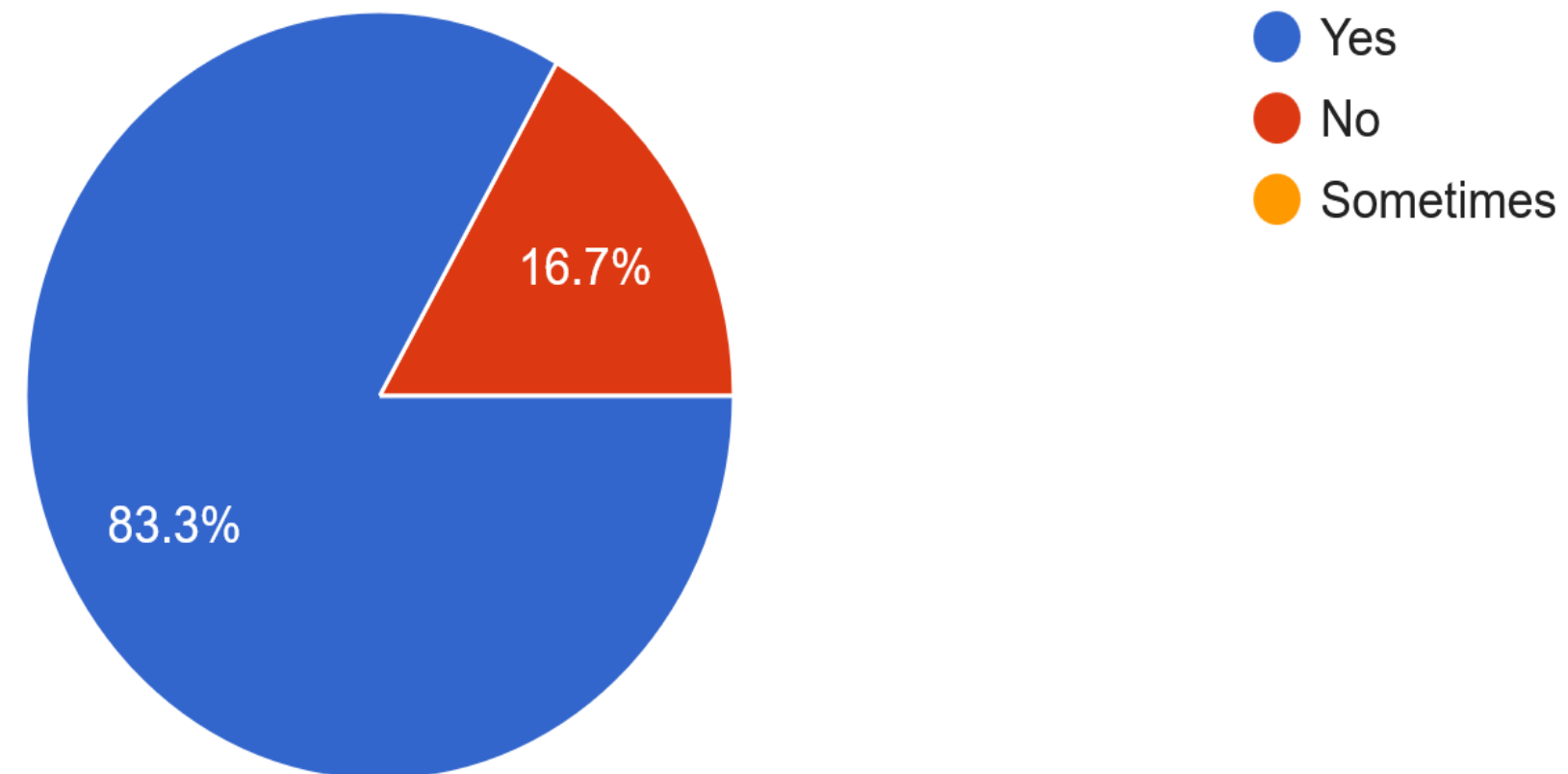
- Programming and Scripting Languages: Python, JavaScript
- Markup and Styling Languages: HTML, CSS
- Detection Model: YOLO
- Libraries: OpenCV, NumPy
- Face Recognition Model: FaceNet
- Similarity Measure: Cosine Similarity
- Database: MySQL
- Tools: VS Code, GitHub

Stakeholder Survey–Overview

- A stakeholder survey was conducted to understand real-world issues in classroom attendance.
- Stakeholders included **faculty members and academic staff**.
- The survey helped identify genuine problems related to **proxy attendance, manual effort, and accuracy** in existing systems.

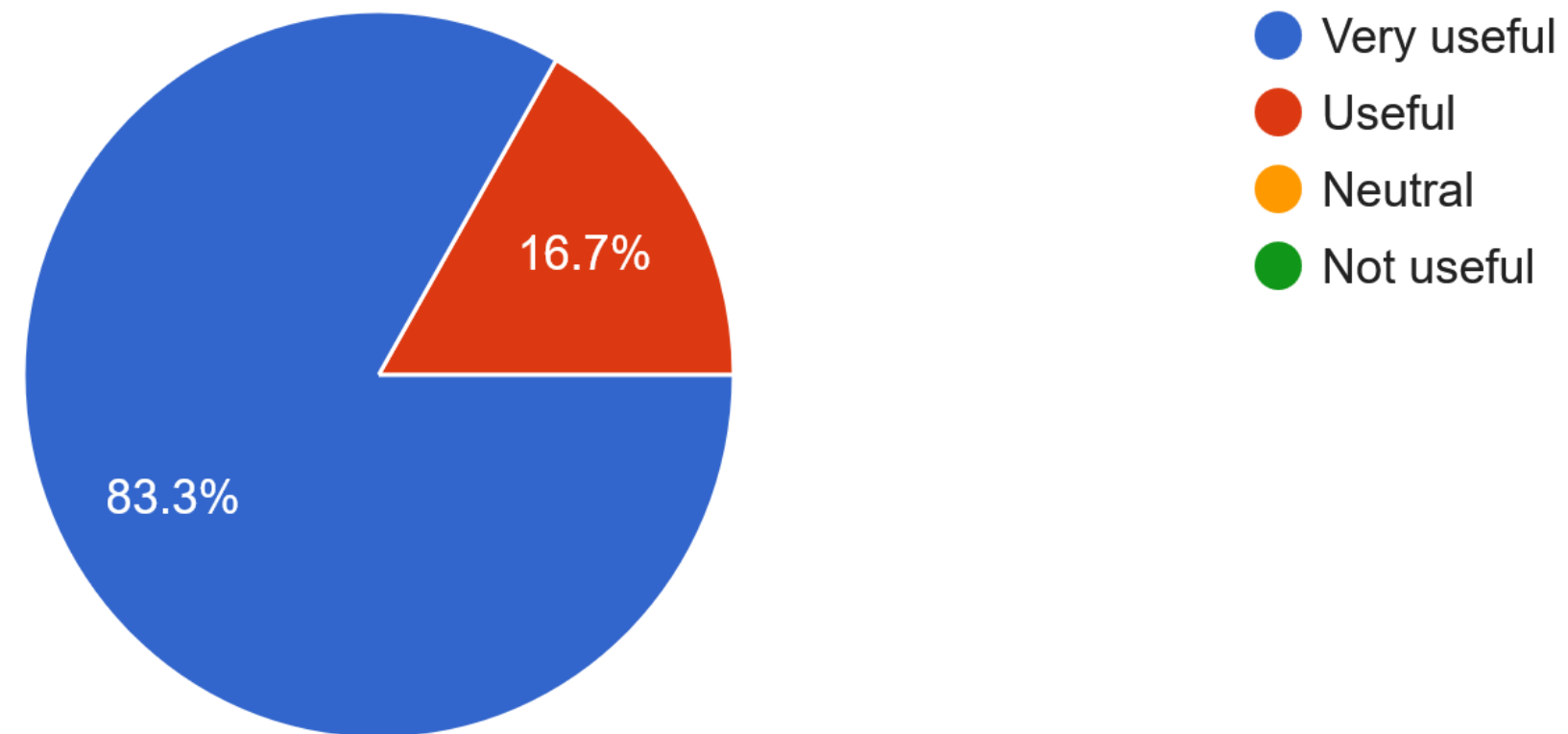
Do you feel the current attendance system is time-consuming?

6 responses



How useful do you think an AI-based Anti-Proxy Attendance System would be?

6 responses



Survey Methodology-Sample

- **Survey Type:** Online questionnaire
- **Tool Used:** Google Forms
- **Number of Respondents:** 6 faculty members
- **Survey Duration:** 7 days

Key Survey Findings

- Majority of faculty reported issues with **proxy and inaccurate attendance.**
- Existing attendance methods are **time-consuming and partially manual.**
- Faculty showed strong interest in an **AI-based automated attendance system.**
- Most respondents believe the system would **improve accuracy, fairness, and transparency**

Problem Identification From Survey

- Stakeholder feedback highlighted **proxy attendance and inefficiency** as critical issues.
- Need for a **secure, accurate, and automated attendance system** was identified.
- Based on survey analysis, the problem statement was finalized as the requirement for an **AI-based Anti-Proxy Classroom Attendance System**.

Work Completed So Far

Core AI System: A complete face recognition pipeline is built using YOLOv8 for detection and FaceNet for accurate student identification.

Camera Integration: The system successfully connects to a live classroom video feed using the Iriun Webcam to capture real-time footage.

Automated Scheduling: A scheduler is implemented to automatically trigger attendance capture according to the configured class timetable.

Database Management: A fully structured MySQL database is set up to reliably store student details, facial embeddings, timetables, and per-period attendance records.

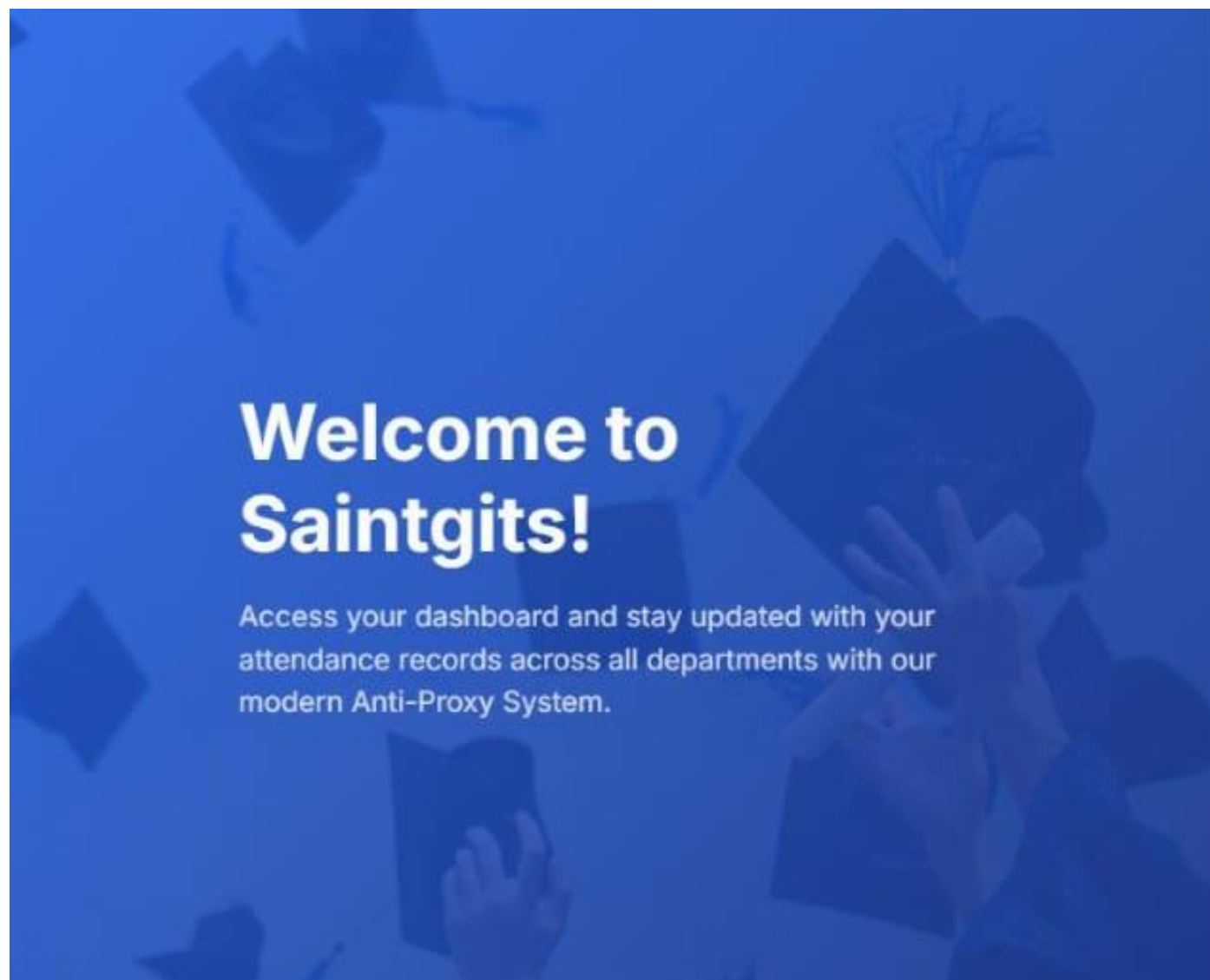
Web Portals: A Flask-based web application is developed featuring three distinct role-based dashboards for Students, Faculty, and Admins.

Key Features: Advanced features like attendance analytics, offline photo uploads for manual marking, and admin override controls are fully integrated.

Data Preparation: Specialized scripts for face alignment, data augmentation, and batch embedding generation are completed to ensure robust AI model training.

Results

Login Page



Sign in to account

Enter your credentials to access your dashboard

Login As

Select Role

Select Role

Student

Faculty

Admin

Password

[Forgot Password?](#)

Sign In →

Admin Dashboard

Admin Dashboard

[Dashboard](#)[Add Faculty](#)[View Classes](#)[Edit Timetable](#)[System Status](#)[Live](#)[Logout](#)

Anti-Proxy Attendance System

[ADMIN](#)[System Admin](#)

Overview

System Administration Dashboard

Total Students

29

Active Classes

1

Total Subjects

18

Turn classroom upload mode on for the whole day

Use this when you want faculty to submit class photos instead of automatic DroidCam captures. Turning it off restores normal DroidCam attendance for the rest of today (the scheduler must be running).

Today: DroidCam auto-capture is ON — normal operation.

[Use classroom photos today \(disable DroidCam\)](#)[Use DroidCam today \(normal\)](#)

Faculty Dashboard

Faculty Dashboard

Dashboard

My Timetable

Logout

Anti-Proxy Attendance System

FACULTY

Dr. Pradeep Kumar P

Welcome, Dr. Pradeep Kumar P

Department of Computer Science

View Class Students

Select a class, date, and period to view attendance details.

Select Date:

29-03-2026



Select Your Period (loads based on date):

-- No classes on this date --



View Attendance

Offline Attendance

Upload a class photo — AI detects faces and marks attendance for the selected period. For **today**, upload works when the admin turns on **full-day classroom photo mode** on the admin dashboard, or if automatic capture logged a camera failure for that period.

Select Date (Max 3 days ago):

29-03-2026



Select Your Period (loads based on date):

-- No classes on this date --



Upload Class Photo:

Choose File

No file chosen

Upload & Detect Faces

Student Dashboard

Student Dashboard

Dashboard

My Timetable

Attendance History

Logout

Saintgits College of Engineering, Kottayam

Student Dashboard

Welcome back! View your academic information and attendance summary.

Student Details

Personal and academic information

Student Name

Joel Garvaziz

Roll Number

CSB-02

Registration Number

REGCSB-02

Batch

B

Semester

6

Email

csb02@university.edu

Phone Number

None

Challenges & Future Scope

Challenges :

1. Webcam Integration:

Difficulty in getting a stable live webcam feed for real-time attendance.

Solution: Changed the webcam software and configured OpenCV settings properly for smooth and stable video capture.

2. Face Data Storage & Matching:

Storing and comparing large FaceNet embedding data in the database was difficult.

Solution: Implemented custom logic to store embeddings and perform accurate face matching.

3. Handling Heavy AI Processing

Heavy AI processing slowed down the web application during attendance processing.

Solution: Separated the AI processing from the web server so the system remains fast and responsive.

Future Scope:

- Liveness detection to prevent proxy attendance using photos/videos
- Automatic face data updating for appearance changes
- Faster face matching using vector databases
- Edge AI deployment for faster processing
- Multi-camera support for large classrooms
- SMS and email alerts for low attendance
- Dedicated Android and iOS mobile applications
- AI-based prediction of academic risk
- Classroom occupancy and usage analytics

Conclusion

The Anti-Proxy Attendance System was successfully developed to provide an automated and secure attendance solution using AI-based face recognition. The system efficiently captures attendance in real time, reduces manual effort, and helps prevent proxy attendance

THANK YOU